

Basic Revision Quiz: Mechanical Oscillations

Target Level: Grade 12 (GS + LS)

Total Marks: 10 pts

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Instructions: Choose the best answer. For calculation questions, clearly show your work in the space provided.

1. A block attached to a horizontal ideal spring performs Simple Harmonic Motion (SHM) along an axis $x'Ox$. The restoring force $\vec{F}$ exerted by the spring on the block of abscissa x is given by the expression: **[0.5 pt]**

- A. $\vec{F} = kx\vec{i}$
- B. $\vec{F} = -kx\vec{i}$
- C. $\vec{F} = mg\vec{i}$
- D. $\vec{F} = -mv\vec{i}$

2. A horizontal elastic pendulum consists of a block of mass $m = 0.1$ kg and a spring of stiffness $k = 10$ N/m. What is the proper angular frequency ω_0 of its free undamped oscillations? (Show your work) **[1.5 pts]**

- A. 10 rad/s
- B. 100 rad/s
- C. 0.1 rad/s
- D. 3.14 rad/s

3. The second-order differential equation governing the un-damped motion of a horizontal elastic pendulum is: **[0.5 pt]**

- A. $x'' - \frac{k}{m}x = 0$
- B. $x'' + \frac{m}{k}x = 0$
- C. $v' + \frac{k}{m}v = 0$
- D. $x'' + \frac{k}{m}x = 0$

4. An oscillator performs SHM with an amplitude $X_m = 5.0$ cm and an angular frequency $\omega_0 = 20$ rad/s. What is the maximum speed V_{max} reached by the mass? (Show your work) **[1.5 pts]**

- A. 100 m/s
- B. 10 m/s
- C. 1.0 m/s
- D. 0.25 m/s

5. At what position during the oscillation is the kinetic energy (KE) of the horizontal pendulum at its maximum value? **[1.0 pt]**
- At the extreme positions ($x = \pm X_m$)
 - Exactly halfway to the amplitude ($x = \pm X_m/2$)
 - At the equilibrium position ($x = 0$)
 - The kinetic energy is constant everywhere
6. A horizontal spring-mass system has a stiffness $k = 200$ N/m. If the maximum elongation of the spring is $X_m = 0.2$ m, what is the total conserved mechanical energy of the system? (Show your work) **[1.5 pts]**
- 40 J
 - 8.0 J
 - 4.0 J
 - 2.0 J
7. In Simple Harmonic Motion, the acceleration a of the center of mass is related to its position x by which of the following equations? **[1.0 pt]**
- $a = -\omega_0 x$
 - $a = -\omega_0^2 x$
 - $a = \omega_0^2 x$
 - $a = -kx^2$
8. The mechanical energy (ME) of a horizontal elastic pendulum is strictly conserved only when: **[0.5 pt]**
- Friction forces are actively acting on the mass.
 - The horizontal surface is perfectly smooth (frictionless).
 - The motion is in the pseudo-periodic regime.
 - The spring reaches its maximum compression.
9. If a weak friction force acts on the oscillating mass, the system enters the "pseudo-periodic" regime. What happens to the mechanical energy and the amplitude over time? **[1.0 pt]**
- Both remain strictly constant.
 - ME remains constant, but the amplitude decreases.
 - ME decreases, but the amplitude remains constant.
 - Both the mechanical energy and the amplitude decrease over time.
10. If the proper period of the position $x(t)$ of an undamped oscillator is $T_0 = 2.0$ s, what is the period of the kinetic energy $KE(t)$? (Show your work) **[1.0 pt]**
- 2.0 s
 - 4.0 s
 - 1.0 s
 - 0.5 s